22 There are only blue pens, green pens and red pens in a box.

The ratio of the number of blue pens to the number of green pens is 2 : 5
The ratio of the number of green pens to the number of red pens is 4 : 1

There are less than 100 pens in the box.

What is the greatest possible number of red pens in the box?

Match the green parts:

Blue : green = 2 : 5 = 8 : 20

Green : red = 4 : 1 = 20 : 5

1 mark
Use the same number of green parts

Blue : green : red = 8 : 20 : 5; total = 33 parts.

3 groups give 99 pens; 4 groups would give 132.

B	G	R
8	20	5
×3 ↓	×3 ↓	×3 ↓
24	60	15

1 mark
Use the largest whole multiplier that keeps the total below 100

1 mark
Maximum red pens = 3 × 5 = 15

15

(Total for Question 22 is 3 marks)